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www.manojmukherjee.co.in (Portfolio)
medium.com/@manojmukherjee777 (Blog)
calendly.com/manojmukherjee777/meet-with-manoj (Other)

Top Skills

GitHub Copilot
Generative AI for Industries
Python (Programming Language)

Languages

English (Professional Working)
Hindi (Professional Working)
Bengali (Native or Bilingual)

Certifications

React: Design Patterns
React: Server-Side Rendering
Learning Next.js
Accessibility for Web Design
Programming Foundations: Web Security

Honors-Awards

University 3rd Rank in BCA(H)
Certificate of Excellence
Excellence in Delivering Krista GA
Star Award #
FS WEST SUPERNOVA

Publications

Case Study of Floating Point Value Problem

Manoj Mukherjee

AI Architect | GenAI & Agentic AI Engineer | RAG • Multi-Agent Systems • FastAPI • AWS/GCP | 10 Years in Enterprise Engineering
Bengaluru, Karnataka, India

Summary

AI Architect & Engineering Leader with 10+ years of experience building and scaling enterprise platforms, modern AI systems, and high-performing engineering teams.I specialize in designing production-grade GenAI architectures, leading end-to-end delivery from concept → POC → production → scale. My work focuses on Agentic RAG, multi-agent orchestration, and AI-native platforms that integrate LLMs, microservices, and cloud infrastructure.Currently driving enterprise AI transformation by:• Architecting Agentic systems (LangGraph, Google ADK, MCP) for complex workflow automation• Building scalable Agentic RAG pipelines with hybrid search, long-term memory, and contextual retrieval• Delivering full-stack AI platforms (FastAPI, Next.js, cloud-native deployments on Kubernetes, openshift, NVIDIA Run.ai)• Defining reusable AI architecture patterns and accelerators for faster adoptionBeyond engineering, I actively operate at the intersection of technology, business, and people:• Lead technical strategy, mentoring engineers and enabling teams to adopt AI-first development• Partner with clients, stakeholders, and leadership to translate business problems into scalable AI solutions• Drive R&D, rapid POCs, and innovation programs to evaluate and onboard emerging AI technologies• Contribute to inner-source and open-source initiatives; recognized with organizational awards for knowledge sharing and impactHands-on across the modern AI ecosystem including LangChain, LangGraph, Google ADK, MCP/ACP/UCP patterns, FastAPI, vector databases, Kubernetes, and enterprise AI platforms (Run:AI, OpenShift AI, Vertex AI, vLLM, Ollama).Previously built AI accelerators for BFSI and contributed to scaling enterprise platforms across cloud, data, and frontend ecosystems.I believe the future belongs to AI-native systems—where agents, data, and services collaborate seamlessly. I focus on building systems and teams that make this real in production.Open to leadership opportunities in AI Architecture, Platform Engineering, and CTO-track roles.My Official Website: https://www.manojmukherjee.co.in

Cypress 10—As Frontend or JavaScript Engineer | by Manoj Mukherjee | Jul, 2022 | Medium

DAMAGED PADDY LEAF DETECTION USING IMAGE PROCESSING

Fibonacci Based Text Hiding Using Image Cryptography

Building a Real-Time AI Agent with LangChain, LangGraph, and Open Source LLMs using Ollama

Experience

Publicis Sapient
Technical Lead
March 2023 - Present (3 years 4 months)
Bengaluru

Architecting enterprise-grade GenAI platforms using Agentic RAG, multi-agent systems, and cloud-native architectures.

Led design and delivery of scalable AI systems enabling enterprise adoption of AI-first solutions.

Key Contributions:
• Built LangGraph-based multi-agent systems with MCP integrations for dynamic AI workflows
• Architected hybrid search using pgvector + MongoDB for improved retrieval and context
• Designed long-term memory and contextual retrieval pipelines
• Developed AI microservices using FastAPI (Python, uv) with containerized deployments
• Built full-stack AI apps with Next.js, TypeScript, and AI SDKs
• Deployed AI workloads on Docker & Kubernetes

Innovation & R&D:
• Built rapid POCs for RFPs and enterprise use cases
• Explored and implemented modern GenAI stack: LangChain, LangGraph, Google ADK, MCP/ACP/UCP, ChatGPT SDK
• Worked with NVIDIA Run:AI, OpenShift AI, Vertex AI, vLLM, Ollama
• Used AI dev tools like GitHub Copilot, Claude Code, Slingshot
• Built systems across cloud, hybrid, and local (Colab) environments

Leadership & Influence:
• Mentored engineers in full-cycle AI engineering (design → deploy → scale)
• Contributed to inner-source & open-source; received Openness Award
• Certified interviewer; supported hiring for AI teams
• Delivered AI workshops and L&D programs across org

Collaboration & Impact:
• Worked with clients, vendors, and partners as AI consultant

• Supported research (PhD/MS), hackathons, and innovation programs
• Recognized by leadership (CEO, CTO) for AI vision and impact

Enabled transition from legacy systems to scalable AI-first architectures.

Kotak Mahindra Bank
Chief Manager (SDE - III)
September 2022 - February 2023 (6 months)
Bengaluru

Served as Chief Manager (SDE - III) at Kotak Mahindra Bank, contributing to the Kotak811 digital banking platform.

Part of the early decision-making and R&D team for the Kotak811 mobile application, working as a React Native developer.

Collaborated on building secure and scalable mobile banking features, with a focus on fraud detection, security, and compliance.

Worked closely with multiple stakeholders and third-party service providers to integrate critical banking services.

Guided external/vendor engineering teams on frontend best practices, architecture, and quality standards.

Participated in product and architecture discussions with senior leadership, including CTO and business stakeholders, to align technology with product vision.

Engaged in UPI 2.0 integration and design discussions, contributing to next-generation digital payment capabilities.

Collaborated cross-functionally with product, backend, and security teams to ensure seamless user experience and system reliability.

Operated in a fast-paced, high-impact environment, contributing to key platform decisions within a short tenure.

A.P. Moller - Maersk
Software Engineer III
August 2021 - August 2022 (1 year 1 month)

Bengaluru

Developed scalable frontend applications using React and Next.js, focusing on performance, reusability, and high-quality user experience.

Worked as a key contributor in establishing microfrontend architecture using Module Federation, enabling scalable and independently deployable frontend modules.

Enabled early adoption of Next.js Module Federation (next-mf) within Maersk applications, helping standardize microfrontend integration patterns across teams.

Contributed to building Server-Side Rendering (SSR) capabilities for microfrontends, improving performance, SEO, and initial load times.

Designed and developed internal frontend tools and engineering utilities to improve developer productivity and consistency across teams.

Played a key role in shaping and evolving the Anchor Design System within Maersk's React ecosystem.

Led code reviews, mentored engineers, and guided best practices in frontend architecture and development workflows.

Collaborated with cross-functional stakeholders (product, backend, design) to deliver scalable and maintainable solutions.

Actively participated in open-source discussions, raised issues, and contributed to improving ecosystem tools.

Unblocked teams by resolving critical technical challenges and providing architectural direction.

Contributed to building systems from scratch, taking ownership from design through delivery.

Krista Software

Senior Software Engineer / Founding Engineer

June 2020 - July 2021 (1 year 2 months)

Bengaluru

• Contributed as a founding member of the core engineering team for an AI-driven automation platform.
• Developed and scaled cross-platform mobile and web applications using the Ionic Framework.
• Led functional development and conducted code reviews to ensure high code quality and performance.
• Awarded the GA Award for outstanding contributions and impact on the engineering team.

Optiv Inc
Software Engineer
June 2019 - June 2020 (1 year 1 month)
Greater Bengaluru Area

Collaborated with the innovation engineering team to build internal platforms using React.js, Redux, GraphQL, Node.js, and Express.

Developed cybersecurity-focused dashboards and internal analytics tools, improving system visibility and operational insights.

Worked on early-stage adoption of GraphQL, contributing to modern API design and efficient data fetching strategies.

Built and managed backend services using Express.js with MongoDB and Mongoose ORM, ensuring scalable and reliable data handling.

Owned full-cycle engineering, including development, testing, CI/CD integration, and production support.

Maintained strong SLA adherence by quickly diagnosing and resolving production issues and bugs.

Shared and drove best practices around state management using Redux, contributing to scalable frontend architecture.

Continuously improved the React ecosystem by enhancing project structure, reusability, and maintainability.

Optimized frontend performance by analyzing and improving Webpack configurations, reducing bundle size and improving load times.

Gained exposure to data engineering concepts using Python and PySpark for large-scale data processing.

Collaborated with open-source communities to unblock development and stay aligned with evolving ecosystem standards.

Engaged with cross-functional stakeholders and engineering leadership (R&D, MD, VP, Head of Engineering) to align technical solutions with business goals.

Hewlett Packard Enterprise
Software Engineer I
December 2018 - June 2019 (7 months)
Bengaluru, Karnataka, India

Contributed to the development of hybrid cloud solutions at Hewlett Packard Enterprise, focusing on frontend platforms for unified cloud service management.

Built user interfaces using React.js and Redux, enabling users to explore and manage cloud services (compute, storage, etc.) from a single platform.

Delivered features for hybrid cloud products such as OpenSphere and GreenLake, ensuring usability, performance, and scalability.

Worked with Node.js services and collaborated with teams building Golang-based microservices, gaining exposure to distributed system design.

Actively contributed to the open-source Grommet design system, improving UI consistency, accessibility, and developer experience.

Participated in SAFe Agile processes, collaborating with cross-functional teams to deliver features in a structured enterprise environment.

Engaged with R&D teams and senior engineering leadership to explore new technologies and align frontend architecture with platform goals.

Gained foundational exposure to cloud and infrastructure concepts including Docker, Kubernetes, virtual machines, and data center operations.

Built a strong understanding of hybrid cloud architecture, focusing on user-centric design and seamless service orchestration.

William O'Neil India
Software Engineer
July 2016 - December 2018 (2 years 6 months)
Bengaluru Area, India

Joined as part of the core engineering team when the organization expanded to India, contributing to building flagship product Panaray, a stock market research and analytics platform.

Developed end-to-end web application features using React.js (from early versions 0.14 to 16.8), delivering real-time market data, analytics, charts, financial insights, and news integration.

Played a key role in setting up the frontend architecture from scratch, including JSX setup, Webpack configuration, and build optimization for performance and scalability.

Designed and implemented state management architecture using Flux, later evolving to Redux with Saga for handling complex asynchronous workflows.

Built backend-for-frontend (BFF) services using Node.js and Express, enabling seamless integration between frontend and backend systems.

Contributed to CI/CD pipeline setup, code reviews, and engineering best practices to ensure high-quality and reliable releases.

Collaborated with cross-functional stakeholders to translate business requirements into scalable technical solutions.

Mentored and trained junior engineers, fostering a strong engineering culture —several mentees progressed to senior and tech lead roles.

Actively shared knowledge through internal tech sessions and discussions, contributing to continuous team growth.

Successfully contributed to the launch and scaling of the Panaray web platform in production.

Education

Visvesvaraya Technological University

Master of Computer Applications (MCA), Computer Programming, Specific Applications · (2013 - 2016)

University of Burdwan

BCA(Hons.), Computer Programming, Specific Applications · (2010 - 2013)

Kamalpur Netaji High School

12th, Higher Secondary · (2008 - 2010)

Kamalpur Netaji High School

10th, Secondary Education · (2006 - 2008)